A TRANSFORMING AND OPTIMIZING VERY LOW-RESOLUTION IMAGE CLASSIFICATION USING HYBRID SPECTRAL-POSE CAPSULE FRAMEWORK

A PROJECT REPORT

Submitted by

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in partial fulfillment forthe award of the degree of

BACHELOR OF TECHNOLOGY

in

COMPUTER SCIENCE AND ENGINEERING

of

FACULTY OF ENGINEERING AND TECHNOLOGY

SRM INSTITUTE OF SCIENCE AND TECHNOLOGY
RAMAPURAM, CHENNAI -600089

May 2026

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Certified that this project report titled **"A TRANSFORMING AND OPTIMIZING VERY LOW-RESOLUTION IMAGE CLASSIFICATION USING HYBRID SPECTRAL-POSE CAPSULE FRAMEWORK"** is the bonafide work of **JAYASRI J [REG.NO: RA2211026020212], MONISHA V [REG.NO: RA2211026020199], RIYA K [REG.NO: RA2211026020223]** who caried out the project work under my supervision. Certified further, that to the best os my knowledge, the work repoted herein does not form any other project report or dissertion on the basis of which a degree or award was conferred on an occasion on this or any other candidate.

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DECLARATION

We hereby declare that the entire work contained in this project report titled **"A TRANSFORMING AND OPTIMIZING VERY LOW-RESOLUTION IMAGE CLASSIFICATION USING HYBRID SPECTRAL-POSE CAPSULE FRAMEWORK"** has been carried out by **JAYASRI J [REG.NO: RA2211026020212], MONISHA V [REG.NO: RA2211026020199], RIYA K [REG.NO: RA2211026020223]** at SRM Institute of Science and Technology, Ramapuram, Chennai-600089, under the guidance of **Dr. VINOTH R, M.E, Ph.D, Assistant Professor**, Department of Computer Science and Engineering.

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ACKNOWLEDGEMENT

We thank the Almighty for giving us the strength and guidance to successfully complete this project. We place on record our deep sense of gratitude to our honourable Chairman **Dr. R. SHIVAKUMAR, MBBS., MD.,** for providing us with the requisite infrastructure throughout the course.

We take the opportunity to extend our hearty and sincere thanks to our **Dean, Dr. M. SAKTHI GANESH., Ph.D.,** for guiding us towards the successful completion of the project.

We take the privilege to extend our hearty and sincere gratitude to the Professor and Head of the department, **Dr. K. RAJA, Ph.D.,** for his suggestions, support and encouragement towards the completion of the project with perfection.

We express our hearty and sincere thanks to our guide **Dr. VINOTH R, M.E,Ph.D., Computer Science and Engineering** for his/her encouragement, motivation and constant guidance throughout this project work.

Our thanks to the teaching and non-teaching staff of the Department of Computer Science and Engineering of SRM Institute of Science and Technology, Chennai, for providing necessary resources for our project.

We also thank our team members for their cooperation, coordination, and support throughout the project. We extend our gratitude to our parents and family members for their continuous support and encouragement

ABSTRACT

Very Low-Resolution (VLR) image classification is a critical area in computer vision with wide-ranging applications from surveillance systems to medical imaging and remote sensing. This project introduces an innovative hybrid framework for improving classification performance on extremely low-resolution images using a Super-Resolution assisted Frequency-Routed Capsule Network (SR + FR-CapsNet). Unlike conventional methods that directly process low-resolution inputs and suffer from loss of important spatial details, our system first enhances image quality using Super-Resolution techniques to restore high-frequency information. The system is structured around a two-stage architecture consisting of an image enhancement module followed by an advanced classification model. When presented with a low-resolution image, the Super-Resolution module reconstructs finer details and improves visual clarity. These enhanced images are then processed by the Frequency-Routed Capsule Network, which captures spatial hierarchies and frequency-domain relationships more effectively than traditional Convolutional Neural Networks. The capsule-based architecture preserves feature orientation and improves the representation of complex patterns in images. This hybrid approach provides significant improvements in classification accuracy and robustness under very low-resolution conditions. The proposed system demonstrates its effectiveness on benchmark datasets such as CIFAR-10 and SVHN, achieving better performance compared to existing CNN and CapsNet-based models. The framework can be applied in real-world scenarios such as surveillance, healthcare diagnostics, and remote sensing, where image quality is often limited.

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LIST OF ACRONYMS AND ABBREVIATIONS

API	APPLICATION PROGRAM INTERFACE
CapsNet	CAPSULE NETWORK
FR-CapsNet	FREQUENCY ROUTED CAPSULE NETWORK
SR	SUPER-RESOLUTION
CIFAR	GRAPHICAL PROCESSING UNIT
SVHN	STREET VIEW HOUSE NUMBERS DATASET

CHAPTER 1 (page number 1

starts from this page)

INTRODUCTION

Very Low-Resolution (VLR) image classification is a crucial aspect of computer vision, especially in applications where images are captured under constrained conditions such as surveillance, remote sensing, and medical imaging. The performance of image classification systems significantly depends on the quality and resolution of input images. However, when images are extremely small, typically of size 8×8 or 16×16 pixels, they contain very limited texture information and reduced discriminative features, making accurate classification a challenging task.Traditional Convolutional Neural Networks (CNNs), which rely heavily on spatial feature extraction, tend to perform poorly under very low-resolution conditions due to the loss of important structural details. Capsule Networks (CapsNets) attempt to address this limitation by preserving spatial hierarchies and relationships between features, but their performance is still constrained when the input images lack sufficient quality and detail.To overcome these challenges, this project proposes a Hybrid Super-Resolution Assisted Frequency-Routed Capsule Network (SR + FR-CapsNet) framework.

1.1 PROBLEM STATEMENT

Edges, textures, and fine-grained details play a vital role in image classification tasks, as they help in distinguishing different objects and patterns within an image. In Very Low-Resolution images, these important features are either highly degraded or completely lost, resulting in blurred structures and weak feature representation. This significantly affects the ability of deep learning models to accurately classify images.Most existing approaches, particularly Convolutional Neural Networks (CNNs), focus primarily on spatial feature extraction and fail to perform effectively when image resolution is extremely low. Although Capsule Networks improve spatial representation by preserving hierarchical relationships, they still depend on the availability of meaningful input features. Super-Resolution techniques have been introduced to enhance image quality, but they are often used independently and not integrated with advanced classification models. Therefore, there is a need for a robust and integrated approach that can enhance image quality while simultaneously improving feature extraction and classification accuracy under very low-resolution conditions.

1.2 AIM OF THE PROJECT

- Enhance classification performance for Very Low-Resolution (VLR) images by improving feature representation and image quality.
- Integrate Super-Resolution techniques to restore high-frequency details and improve visual clarity of low-resolution inputs.
- Develop a Frequency-Routed Capsule Network to preserve spatial hierarchies and capture important feature relationships.
- Improve model robustness and generalization across different datasets and varying resolution conditions.
- Optimize overall system performance to achieve higher accuracy with efficient computation.

1.3 PROJECT DOMAIN

The domain of this project is Deep Learning (DL) and Computer Vision, particularly focusing on image classification using advanced neural network architectures such as Super-Resolution models and Capsule Networks for processing Very Low-Resolution images.

1.4 SCOPE OF THE PROJECT

The scope of this project encompasses the development and implementation of a hybrid deep learning framework for Very Low-Resolution image classification. This includes modules for image preprocessing, Super-Resolution enhancement, feature extraction, and classification using Frequency-Routed Capsule Networks. The system focuses on improving feature representation and classification accuracy under low-resolution conditions. The project aims to address challenges such as loss of spatial details, weak feature extraction, and poor model performance by integrating image enhancement and advanced learning techniques. The proposed system can be extended to real-world applications such as surveillance, medical imaging, and remote sensing.

1.5 METHODOLOGY

In this project, a Hybrid Super-Resolution Assisted Frequency-Routed Capsule Network framework is implemented to improve the classification performance of Very Low-Resolution images. The methodology begins with data preprocessing, where benchmark datasets such as CIFAR-10 and SVHN are collected and resized to simulate low-resolution conditions such as 8×8 and 16×16 pixels. Data normalization and augmentation techniques are applied to enhance model generalization and robustness.The Super-Resolution module is then used to enhance the degraded low-resolution images by restoring high-frequency details and improving visual clarity. The enhanced images are passed through convolutional layers for extracting important spatial features such as edges, textures, and structural patterns. These extracted features are further processed using a Frequency-Routed Capsule Network, which preserves spatial hierarchies and captures frequency-domain relationships for better feature representation.

1.6 ORGANIZATION OF THE REPORT

Chapter 2 contains the literature review of relevant research papers related to Very Low-Resolution image classification, Super-Resolution techniques, and Capsule Networks. It provides an overview of existing approaches, their advantages, limitations, and the need for an improved hybrid framework. Various state-of-the-art methods are analyzed to understand their performance under low-resolution conditions. The chapter also highlights research gaps and motivates the development of the proposed system.

Chapter 3 describes the existing system and highlights the challenges faced in traditional Convolutional Neural Networks and Capsule Network-based approaches for Very Low-Resolution image classification. It also presents the proposed system, which integrates Super-Resolution enhancement with Frequency-Routed Capsule Networks to improve performance and accuracy. A comparison between existing and proposed approaches is discussed to justify the improvements. The advantages and limitations of the proposed system are also explained.

Chapter 4 explains the proposed work in detail, including the overall system architecture and block diagram. It describes the complete image processing pipeline consisting of Super-Resolution enhancement, feature extraction, and classification modules. The working of each modle is explained clearly along with the data flow between them. Diagrams such as block diagrams and workflow representations are included for better understanding of the system.

Chapter 5 focuses on the implementation and testing of the proposed system. It includes details about dataset preparation, preprocessing techniques, model development, and training procedures. Various testing methods such as unit testing, integration testing, and performance evaluation are carried out to ensure system reliability. The tools and technologies used for implementation are also discussed in this chapter. The outputs and intermediate results of the system are illustrated using simple images and visual representations, as shown in Figures 5.1 and 5.2.

Chapter 6 presents the results and discussion of the proposed model. It analyzes the performance using evaluation metrics such as accuracy, precision, recall, and F1-score. The results are compared with existing methods to demonstrate the improvements achieved. Graphs and charts are used to visually represent the performance of the system under different conditions. The experimental results and performance comparisons are illustrated using graphical representations, as shown in Figures 6.1 and 6.2.

Chapter 7 provides the conclusion and outlines future enhancements. It summarizes the effectiveness of the proposed system and highlights the key contributions of the project. The limitations of the current system are discussed, along with possible improvements. Future work may include optimization techniques, advanced architectures, and real-time deployment.

Chapter 8 contains the source code details and additional information related to the project implementation. It may also include sample outputs, screenshots, and documentation required for understanding the system. This chapter serves as a reference for further development and practical usage of the system.

CHAPTER 2

LITERATURE REVIEW

Very Low-Resolution (VLR) image classification remains a challenging problem due to the limited availability of discriminative features in small-sized images. Zou et al. [1] emphasized that as image resolution decreases, important structural and texture details are lost, making feature extraction difficult. Their work highlighted the importance of designing models that can effectively handle degraded visual information without relying on high-resolution inputs.

Capsule Networks, introduced by Sabour et al. [2], attempt to overcome the limitations of traditional CNNs by preserving spatial hierarchies and relationships between features. The concept of dynamic routing allows capsules to encode orientation and positional information more effectively. However, Capsule Networks require high computational power and are sensitive to noisy inputs, which limits their performance in very low-resolution scenarios. Koziarski et al. [3] conducted a detailed analysis of deep learning models under low-resolution constraints and observed a significant drop in classification accuracy. Their findings revealed that standard CNN architectures are not well-suited for extremely small images due to insufficient feature representation. This work motivated further research into specialized architectures for handling VLR data. Singh et al. [4] proposed a Dual Directed Capsule Network to improve the feature extraction process. Their model introduced enhanced routing strategies to improve classification performance on low-resolution datasets. While the results showed improvement, the complexity of the model increased, making it less efficient for practical deployment.

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Super-Resolution (SR) techniques play a crucial role in improving image quality by reconstructing high-frequency details. Wang et al. [5] introduced ESRGAN, which significantly improves perceptual quality and generates sharper images. The model uses adversarial learning to produce realistic textures. However, its high computational cost and training complexity limit its real-time applicability.

Liang et al. [6] proposed SwinIR, a transformer-based Super-Resolution model that achieves excellent performance in image restoration tasks. By leveraging self-attention mechanisms, the model captures long-range dependencies in images. Despite its effectiveness, the requirement for large computational resources makes it challenging to implement in resource-constrained environments.

Dewasurendra et al. [7] introduced Frequency-Routed Capsule Networks (FR-CapsNet), which incorporate frequency-domain information into capsule routing. This approach improves feature representation by combining spatial and frequency features. However, the model does not address the issue of poor image quality in Very Low-Resolution inputs. Lim et al. [8] developed Enhanced Deep Residual Networks (EDSR) for Super-Resolution tasks. Their model improves image reconstruction by using deep residual learning techniques. While EDSR produces high-quality images, it focuses only on enhancement and does not contribute directly to classification accuracy. Furthermore, recent studies have explored the use of Vision Transformers (ViT) for image classification tasks. Dosovitskiy et al. [9] introduced the Vision Transformer model, which utilizes self-attention mechanisms to capture global dependencies within images. While ViT achieves strong performance on high-resolution datasets, its effectiveness decreases when applied to Very Low-Resolution images due to the lack of sufficient visual information.

To address this limitation, hybrid models combining Convolutional Neural Networks and transformer architectures have been proposed. These models leverage the local feature extraction capability of CNNs along with the global context modeling of transformers. However, such hybrid architectures often introduce additional computational complexity and require large-scale datasets for effective training [10]. Another important direction in research involves the use of Generative Adversarial Networks (GANs) for enhancing low-resolution images. Ledig et al. [11] proposed SRGAN, which generates high-resolution images with realistic textures using adversarial learning. Although GAN-based approaches produce visually appealing results, they may introduce artifacts and instability during training, which can affect the reliability of classification system

In addition, researchers have investigated the role of frequency-domain analysis in improving feature representation. Frequency-based methods focus on capturing both low-frequency and high-frequency components of an image, which are essential for preserving structural information. These approaches have shown promising results in enhancing classification performance, particularly when combined with deep learning models [12].

Moreover, ensemble learning techniques have been explored to improve classification accuracy by combining multiple models. Ensemble methods reduce model variance and improve robustness; however, they significantly increase computational cost and system complexity, making them less suitable for real-time applications [13]. Recent advancements also include the use of meta-learning and self-supervised learning techniques to improve model generalization. These approaches enable models to learn from limited data and adapt to new tasks more effectively. Despite their advantages, they still face challenges in handling extremely low-resolution images where feature information is severely limited [14]. In addition, graph-based learning approaches have been explored to model relationships between different features in an image. Graph Neural Networks (GNNs) help in capturing complex feature dependencies and improving representation learning. However, their effectiveness in Very Low-Resolution image classification is limited due to the insufficient availability of meaningful node features [15].

Another emerging area involves the use of contrastive learning techniques for improving feature discrimination. Contrastive learning enables models to learn better representations by distinguishing between similar and dissimilar image pairs. While this approach improves robustness, its performance is still constrained when applied to extremely low-resolution images with limited visual information [16]. Researchers have also investigated the use of multi-task learning frameworks, where a model is trained to perform multiple related tasks simultaneously, such as image enhancement and classification. This approach improves overall learning efficiency and feature sharing, but it increases model complexity and requires careful optimization to balance different objectives [17]. Furthermore, attention-based capsule networks have been proposed to enhance feature selection by focusing on the most relevant parts of the image. These models combine the strengths of attention mechanisms and capsule architectures to improve classification accuracy. However, they still rely on the quality of input images and may not perform effectively when the resolution is extremely low [18].

From the extended literature review, it is evident that although various techniques such as CNNs, Capsule Networks, Transformers, and GANs have been proposed, each approach has its own limitations when applied to Very Low-Resolution image classification. Most existing systems either focus on improving image quality or enhancing classification performance independently, without effectively combining both aspects.

Therefore, the proposed Hybrid Super-Resolution Assisted Frequency-Routed Capsule Network framework aims to bridge this gap by integrating Super-Resolution enhancement with advanced feature extraction and classification mechanisms. This approach is expected to improve both image quality and classification accuracy while maintaining computational efficiency.

CHAPTER 3

PROJECT DESCRIPTION

3.1 EXISTING SYSTEM

The existing approaches for image classification, particularly under Very Low-Resolution (VLR) conditions, mainly rely on Convolutional Neural Networks (CNNs). These models focus on extracting spatial features such as edges, textures, and patterns from input images. However, when the resolution of images is extremely low, important visual details are lost, leading to poor feature representation and reduced classification accuracy. Capsule Networks were introduced to overcome some limitations of CNNs by preserving spatial hierarchies and relationships between features. Although they improve representation learning, their performance is still dependent on the quality of input images. In Very Low-Resolution scenarios, the lack of sufficient feature information limits their effectiveness. Super-Resolution techniques have been proposed to enhance image quality by reconstructing high-resolution images from low-resolution inputs. However, in most existing systems, Super-Resolution and classification are treated as separate processes, resulting in inefficient feature learning. These limitations highlight the need for an integrated approach that can simultaneously enhance image quality and improve classification performance.

3.2 PROPOSED SYSTEM

In In the proposed system, we address the challenges of Very Low-Resolution image classification by introducing a Hybrid Super-Resolution Assisted Frequency-Routed Capsule Network framework. This approach integrates image enhancement and classification into a single unified model to improve overall system performance. The system begins with preprocessing, where input images are resized and normalized to simulate low-resolution conditions. A Super-Resolution module is then applied to enhance image quality by restoring high-frequency details and improving visual clarity. The enhanced images are passed through convolutional layers for extracting meaningful spatial features. These extracted features are further processed using a Frequency-Routed Capsule Network, which preserves spatial hierarchies and incorporates frequency-domain information for better feature representation. This combination enables the model to effectively classify images even under extremely low-resolution conditions.

3.2.1 ADVANTAGES

- Improve classification accuracy for very low resolution images.
- Enhances image quality using super resolution techniques.
- Preserves spatial relationships using capsule networks.
- Incorporates frequency-domain information for better feature extraction.
- Reduces information loss and improves feature representation.
- Provides better generalization across different datasets.

3.3 FEASIBILITY STUDY

A feasibility study is conducted to evaluate the practicality and effectiveness of the proposed system. The feasibility analysis is carried out across three main dimensions:

- Economic Feasibility
- Technical Feasibility
- Social Feasibility

3.3.1 ECONOMIC FEASIBILITY

The proposed system does not require expensive hardware or specialized equipment, making it economically feasible. The development is carried out using freely available tools and open-source libraries, reducing overall cost. Therefore, the system can be implemented without significant financial investment.

3.3.2 TECHNICAL FEASIBILITY

The proposed system is based on deep learning techniques using tools such as Python, TensorFlow, and Jupyter Notebook. These tools are widely available and easy to use, ensuring technical feasibility. The required computational resources are manageable, and the system can be implemented efficiently with standard hardware configurations.

3.3.3 SOCIAL FEASIBILITY

The proposed system is user-friendly and does not require advanced technical knowledge for operation. It can be applied in various real-world domains such as surveillance, healthcare, and security systems. The system improves decision-making and enhances performance, making it socially beneficial.

3.4 SYSTEM SPECIFICATION

An effective system requires proper hardware and software components to ensure smooth execution. The system specifications define the necessary requirements for implementing the proposed model efficiently.

3.4.1 HARDWARE SPECIFICATION

- Processor: Intel i5, i7 or higher.
- RAM: Minimum 8 GB (Recommended 16 GB or above)
- Storage: Minimum 100 GB.
- Internet: Required for dataset download and updates.

3.4.2 SOFTWARE SPECIFICATION

- Python
- Anaconda
- Jupyter Notebook
- TensorFlow
- Keras
- opencv-python
- pandas
- matplotlib

CHAPTER 4

PROPOSED WORK

4.1 GENERAL ARCHITECTURE

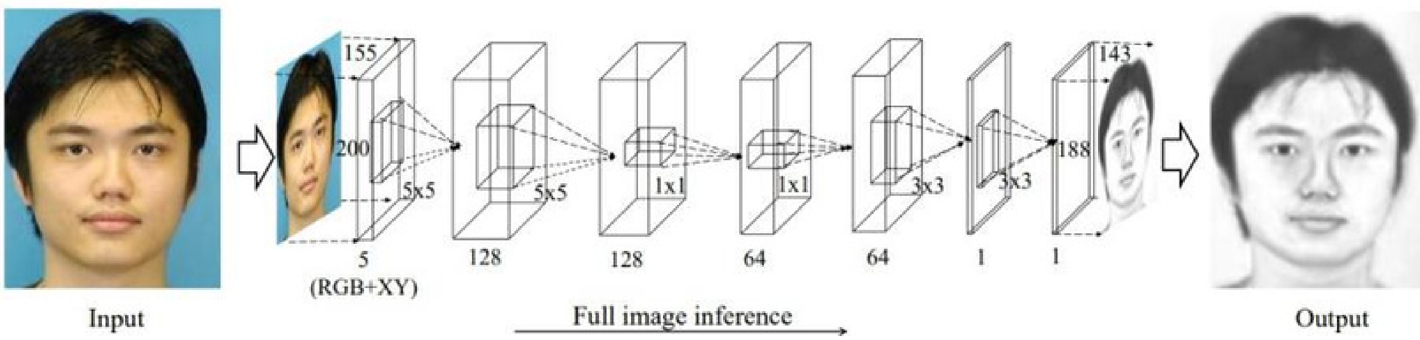

Figure 4.1: **Architecture Diagram**

Figure 4.1 illustrates a potential architecture inspired by Collaborative Generative Representation Learning Neural Network (CGRL-NN), tailored for a facial recognition system.

4.2 DESIGN PHASE

During the design phase, diverse diagrams and models are crafted to depict various elements of the system, such as its components, interactions, and data flow. These diagrams, including UML, sequence, use case, and data flow diagrams, aid in conveying the system's design and functionality to stakeholders and development teams. In essence, the design phase is pivotal for ensuring that the software solution achieves its objectives in a proficient and effective manner.

4.2.1 DATA FLOW DIAGRAM

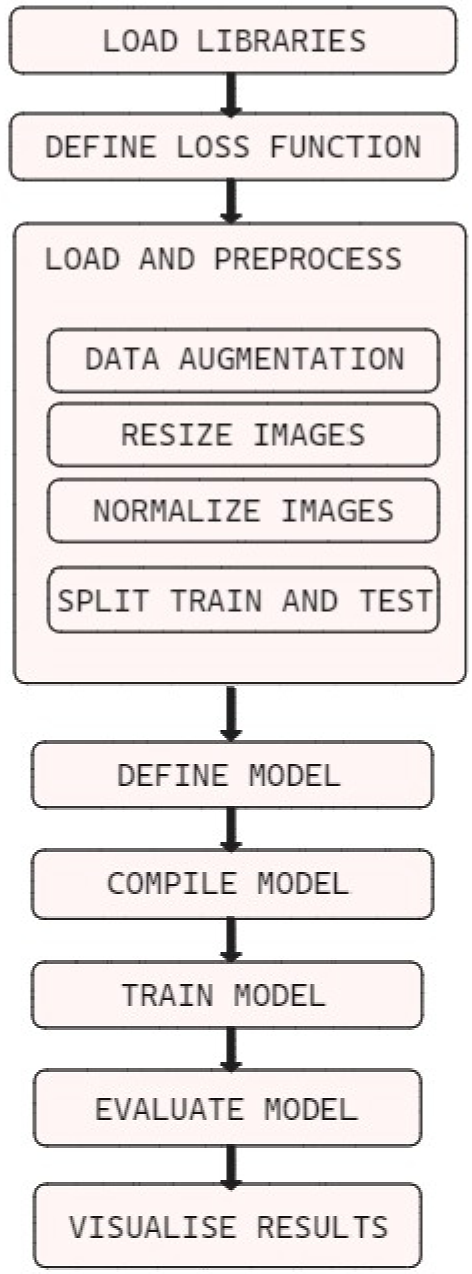

Figure 4.2: **Data Flow Diagram**

This Figure 4.2 shows a data flow diagram of theimage processing pipeline. Raw image data enters first, followed by pre-processing to enhance quality. Feature extraction then condenses the data by selecting key characteristics. Finally, a fully connected neural network layer analyzes these features for image classification or prediction, outputting the processed data.

4.2.2 UML DIAGRAM

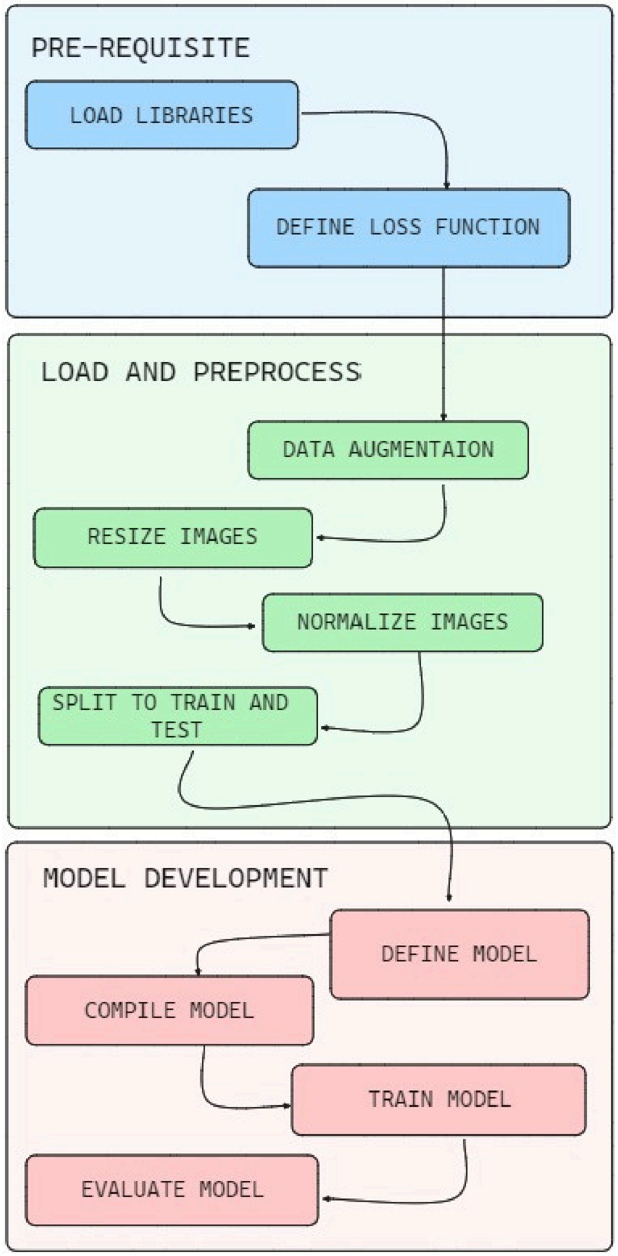

Figure 4.3: **UML Diagram**

The Figure 4.3 is a UML diagram depicting the image processing pipeline. It consists of three main stages: image preprocessing, feature extraction, and sketch to image prediction. Preprocessing aims to improve the image data by suppressing noise or enhancing features. Feature extraction reduces the data by selecting the most relevant information. Finally, sketch to image prediction utilizes a neural network to convert a sketch into a complete image.

4.2.3 USE CASE DIAGRAM

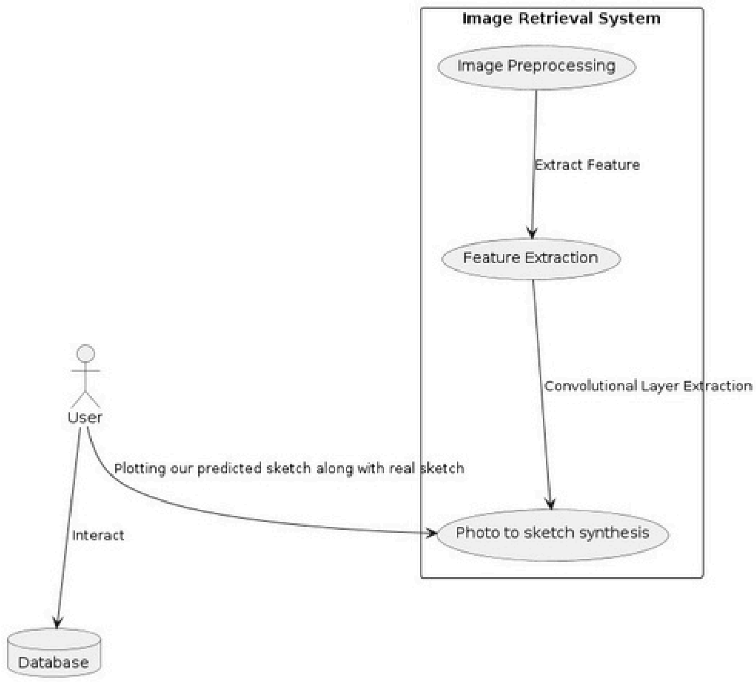

Figure 4.4: **Use Case Diagram**

The diagram in Figure 4.4 illustrates the flow of the Collaborative Generative Representation Learning technique for mapping sketches to images, involving Image Preprocessing, Feature Extraction, and Sketch to Image Prediction.

4.2.4 SEQUENCE DIAGRAM

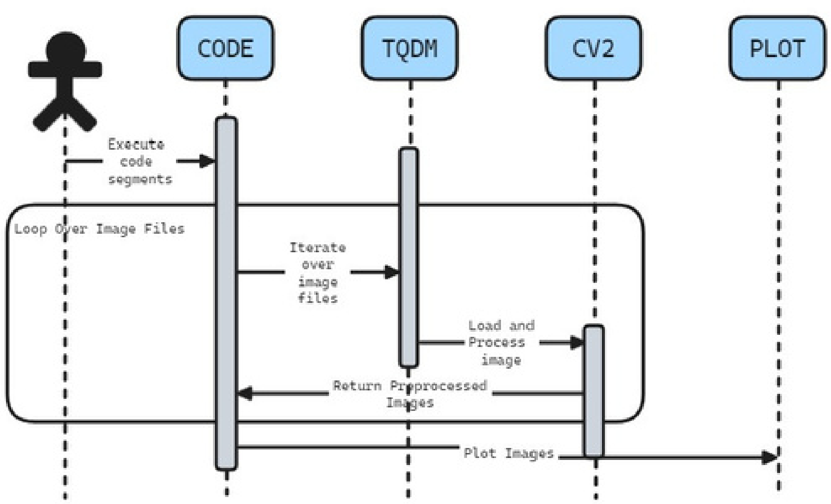

Figure 4.5: **Sequence Diagram**

The sequence diagram illustrates the process of loading and preprocessing images from a dataset directory. Initiated by the user, the code module iterates over image files with progress tracking by the tqdm library. Using the OpenCV library, images are loaded, preprocessed, and returned to the code module. Matplotlib then visualizes the preprocessed images. This iterative process continues until all images are processed, concluding the sequence.

4.3 MODULE DESCRIPTION

The following modules form the core components of our image processing and analysis pipeline, each serving a distinct yet interconnected role in Photo - Sketch synthesis :

4.3.1 MODULE1 : IMAGE PREPROCESSING

Pre-processing involves applying various procedures to images at their most basic level, where both input and output are represented as intensity images. These images closely resemble the original data captured by sensors, typically appearing as matrices of brightness values. The main aim of pre-processing is to refine image data, reducing unintended distortions while enhancing specific attributes vital for subsequent processing.

4.3.2 MODULE 2 : FEATURE EXTRACTION

Feature extraction constitutes a pivotal component of the dimensionality reduction process, wherein an initial set of raw data undergoes segmentation and condensation into more manageable groups, facilitating streamlined processing. A defining attribute of these extensive data sets is their abundance of variables, demanding significant computational resources for analysis. Feature extraction addresses this challenge by discerning and amalgamating variables into features, thereby substantively diminishing the data volume. These resultant features are conducive to efficient processing while retaining the capacity to accurately depict the original data set with fidelity and precision.

4.3.3 MODULE 3: IMAGE TO SKETCH SYNTHESIS

The neural network is regarded as an adept feature extractor, comprising two primary components integral to its functioning. Firstly, the feature extractor incorporates convolutional and pooling layers, tasked with autonomously discerning and assimilating key attributes from raw data. Subsequently, the fully connected layer employs the acquired features to execute classification tasks.The input layer serves to ingest individual data values, while the output layer yields results corresponding to the number of distinct categories requiring classification.Within the convolutional layer, localized regions of the data undergo scrutiny to extract pertinent features, while pooling layers serve to streamline computational complexity by reducing parameter quantities.

4.3.4 STEP 2: PROCESSING OF DATA

- The CUHK (CUFS) dataset contains 188 images of students from the Chinese University of Hong Kong along with the sketches of those images [54].
- The dataset is assigned to image_path and sketch_path and sorted in alphanumeric order.
- In Figure 4.7, using the opencv library, they are color corrected, resized, rotated and converted from an image to an array.

```
for i in tqdm(image_file):
    image = cv2.imread(image_path + '/' + i,1)
    image = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)
    image = cv2.resize(image, (SIZE, SIZE))
    image = image.astype('float32') / 255.0
    img_array.append(img_to_array(image))
    img1 = cv2.flip(image,1)
    img_array.append(img_to_array(img1))
    img2 = cv2.flip(image,-1)
    img_array.append(img_to_array(img2))
    img3 = cv2.flip(image,-1)
    img3 = cv2.flip(img3,1)
    img_array.append(img_to_array(img3))
    img4 = cv2.rotate(image, cv2.ROTATE_90_CLOCKWISE)
    img_array.append(img_to_array(img4))
    img5 = cv2.flip(img4,1)
    img_array.append(img_to_array(img5))
    img6 = cv2.rotate(image, cv2.ROTATE_90_COUNTERCLOCKWISE)
    img_array.append(img_to_array(img6))
    img7 = cv2.flip(img6,1)
```

Figure 4.6: **Pre-processing Of Data**

4.3.5 STEP:3 SPLIT THE DATA

- After the preprocessing part, both sketches and images are split to a training and testing split.
- The training data contains 80 percent of the database. This is applicable for both the images and the sketch version of the images
- The test data contains the rest 20 percent of the database.

4.3.6 DATASETS SAMPLE

Figure 4.7: **Dataset Of Photos**

Figure 4.8: **Dataset Of Sketches**

4.3.7 STEP 4: BUILDING THE MODEL

Ourmodel ofchoiceisanencoder-decoder architecture-based model implemented using the Keras functional API.

INPUT LAYER:

Theinput layerofthe model is defined with the input shape being (SIZE, SIZE, 3) denoting an image with dimensions ‘size x size’ and 3 channels (RGB)

ENCODER SECTION: Weproceed to the encoder section by implementing a downsample function that reduces the spatial dimensions of the input tensor with an increase in depth. Each downsample operation uses a 4x4 convolutional layer with a specified number of filters.

DECODER SECTION: The decoder sectionmirrors the encoder but in reverse. This is done by using upsample operations to increase spatial dimensions while decreasing depth .Each upsample operation is paired with a 4x4 transposed convolutional layer (Conv2DTranspose) except for the last two layers that use a 2x2 transposed convolutional layer.

MODEL CREATION:

Finally, the encoderinput and decoder output is used to create a Keras model object. **Figure 4.10** below shows the model summary.

Layer (type)	Output Shape	Param #
input_layer (InputLayer)	(None, 256, 256, 3)	0
sequential (Sequential)	(None, 127, 127, 16)	768
sequential_1 (Sequential)	(None, 62, 62, 32)	8,320
sequential_2 (Sequential)	(None, 30, 30, 64)	32,768
sequential_3 (Sequential)	(None, 14, 14, 128)	131,584
sequential_4 (Sequential)	(None, 6, 6, 256)	525,312
sequential_5 (Sequential)	(None, 2, 2, 512)	2,099,200
sequential_6 (Sequential)	(None, 6, 6, 512)	4,194,304
sequential_7 (Sequential)	(None, 14, 14, 256)	2,097,152
sequential_8 (Sequential)	(None, 30, 30, 128)	524,288
sequential_9 (Sequential)	(None, 62, 62, 64)	131,072
sequential_10 (Sequential)	(None, 126, 126, 32)	32,768
sequential_11 (Sequential)	(None, 254, 254, 16)	8,192
conv2d_transpose_6 (Conv2DTranspose)	(None, 255, 255, 8)	520
conv2d_transpose_7 (Conv2DTranspose)	(None, 256, 256, 3)	99

Figure 4.9**: Model Summary**

4.3.8 STEP 5: COMPILING AND TRAINING THE MODEL

The model is now compiled by using the Adam optimizer with a specified learning rate of 0.0001. The model's performance during training and evaluation is monitored using the accuracy metric ('acc'). The compiled model is now made to fit or train using the training data. The epochs in this case have been set to 150 as this showed us the best results. **Figure 4.11** shows the lines of code used to compile and fit the model.

```
model.compile(optimizer = Adam(learning_rate = 0.001), loss = 'mean_absolute_error',
              metrics = ['acc'])
```

```
model.fit(train_image, train_sketch_image, epochs = 150, verbose = 0)
```

Figure 4.10: **Code To Compile Model And Fit Model To Dataset**

CHAPTER 5

IMPLEMENTATION AND TESTING

5.1 INPUT AND OUTPUT

5.1.1 IMAGE OF THE SUBJECT

- A random student has been selected from the 188 students for evaluation of the model.
- This image is fed into the Neural Network and the network gives a sketch synthesized output of what the predicted sketch would look like for the input image.
- **Figure 5.1** below shows the original, augmented image of the subject

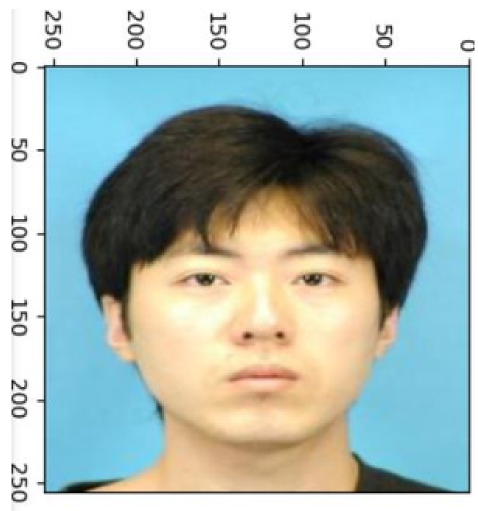

Figure 5.1: **Original Image Of The Subject**

5.1.2 PREDICTED SKETCH SYNTHESIS OFTHE SUBJECT

Figure 5.2 below shows the sketch synthesized output by the neural network

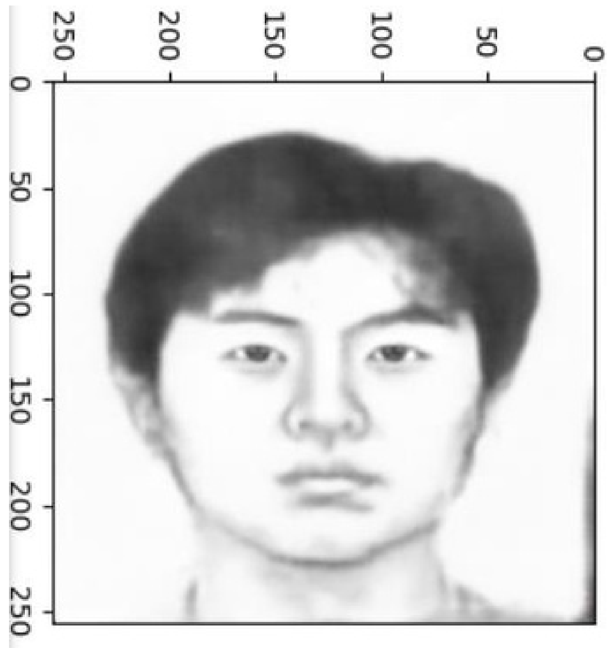

Figure 5.2 **:Predicted Sketch Of The Subject**

5.2 TESTING

In photo-sketch synthesis, testing serves to validate whether the generated sketches accurately depict the original photos. This ensures that the autoencoder-based generative model conforms to specified criteria. Testing is conducted to confirm whether the synthesized sketches successfully capture essential details and uphold fidelity to the source images. The process entails rigorous examination aimed at verifying that the model effectively achieves its intended purpose.

5.2.1 TYPES OF TESTING

5.2.2 UNIT TESTING

Unit testing is a beneficiable software testing method where the units of source code is tested to check the efficiency and correctness of the program. **Figure 5.3** below contains the code for the image augmentation.

INPUT:

```
for i in tqdm(image_file):
    image = cv2.imread(image_path + '/' + i,1)
    image = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)
    image = cv2.resize(image, (SIZE, SIZE))
    image = image.astype('float32') / 255.0
    img_array.append(img_to_array(image))
    img1 = cv2.flip(image,1)
    img_array.append(img_to_array(img1))
    img2 = cv2.flip(image,-1)
    img_array.append(img_to_array(img2))
    img3 = cv2.flip(image,-1)
    img3 = cv2.flip(img3,1)
    img_array.append(img_to_array(img3))
```

```
100%|██████████████████████████████████████████████████| 188/188 [00:02<00:00, 75.11it/s]
```

```
for i in tqdm(sketch_file):
    image = cv2.imread(sketch_path + '/' + i,1)
    image = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)
    image = cv2.resize(image, (SIZE, SIZE))
    image = image.astype('float32') / 255.0
    sketch_array.append(img_to_array(image))
    img1 = cv2.flip(image,1)
    sketch_array.append(img_to_array(img1))
    img2 = cv2.flip(image,-1)
    sketch_array.append(img_to_array(img2))
    img3 = cv2.flip(image,-1)
    img3 = cv2.flip(img3,1)
    sketch_array.append(img_to_array(img3))
```

```
100%|██████████████████████████████████████████████████| 188/188 [00:02<00:00, 80.51it/s]
```

Figure 5.3:**Image Augmentation Using opencv-python Library**

TEST RESULT

- Images of size 224*224 pixels are considered
- The images undergo augmentation using opencv-python
- The considered images are loaded into an array for preprocessing.

5.2.3 INTEGRATION TESTING

INPUT:

Figure 5.4 below showsthecodesnippet for printing the images and sketches while **Figure 5.5** shows theoutputofthe code

```
def plot_images(image, sketches):
    plt.figure(figsize=(7,7))
    plt.subplot(1,2,1)
    plt.title('Sketches ', color = 'black', fontsize = 20)
    plt.imshow(sketches)
    plt.subplot(1,2,2)
    plt.title('Image', color = 'green', fontsize = 20)
    plt.imshow(image)


    plt.show()
```

```
ls = [i for i in range(0,65,8)]
for i in ls:
    plot_images(img_array[i],sketch_array[i])
```

Figure 5.4: **Code For Printing Images And Their Corresponding Sketches**

TEST RESULT

- A The images from the Images folder and their corresponding sketches from the sketches folder are loaded
- The images and the sketches are then made to display as output parallelly using the matplotlib library.

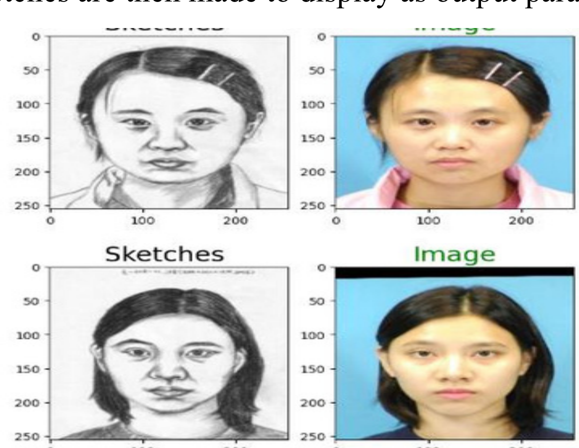

Figure 5.5: **Sketches And Their Image Printed As Output**

5.2.4 FUNCTIONAL TESTING

INPUT

```
[50]: train_sketch_image = sketch_array[:1000]
      train_image = img_array[:1000]

[51]: test_sketch_image = sketch_array[1000:]
      test_image = img_array[1000:]

[52]: train_sketch_image = np.reshape(train_sketch_image,(len(train_sketch_image),SIZE,SIZE,3))

[53]: train_image = np.reshape(train_image, (len(train_image),SIZE,SIZE,3))

[54]: print('Train color image shape:',train_image.shape)

      Train color image shape: (1000, 256, 256, 3)

[55]: test_sketch_image = np.reshape(test_sketch_image,(len(test_sketch_image),SIZE,SIZE,3))

[56]: test_image = np.reshape(test_image, (len(test_image),SIZE,SIZE,3))

[57]: print('Test color image shape',test_image.shape)

      Test color image shape (504, 256, 256, 3)
```

Figure 5.6: **Training And Testing Split**

```
[62]: model.compile(optimizer = Adam(learning_rate = 0.001), loss = 'mean_absolute_error',
                    metrics = ['acc'])

[69]: model.fit(train_image, train_sketch_image, epochs = 150, verbose = 0)

[69]: <keras.src.callbacks.history.History at 0x18b6b4f09d0>

[70]: prediction_on_test_data = model.evaluate(test_image, test_sketch_image)
      print("Loss: ", prediction_on_test_data[0])
      print("Accuracy: ", np.round(prediction_on_test_data[1] * 100,1))

      16/16 ━━━━━━━━━━━━━━━━━━━━ 2s 134ms/step - acc: 0.3984 - loss: 0.0797
      Loss:  0.08201704174280167
      Accuracy:  39.7
```

Figure 5.7: **Model Is Being Complied And Trained**

TEST RESULT

- All the images from data sets are loaded into the model and carried out for training.

- Training is done by considering each image and saving the characteristics of the image

CHAPTER 6
RESULTS AND DISCUSSIONS

6.1 EFFICIENCY OF THE PROPOSED SYSTEM

The current proposed system uses an encoder-decoder architecture to learn a mapping between images and sketches. Autoencoders having a simple but efficient architecture results in faster training times and lower computational requirements. It also does not require Labeled data as it trains in an unsupervised manner. The current model gives us an accuracy of about 40 percent. With the Autoencoder's generative capabilities, this is the right neural network for image generation tasks

6.2 COMPARISON OF EXISTING AND PROPOSED SYSTEM

The proposed system can achieve good performance with relatively smaller datasets compared to more data-hungry models, reducing data requirements and training time. It is also well-suited for tasks such as image denoising, image compression, anomaly detection, and image-to-image translation, providing versatility in application areas. The proposed system can also generalize well to unseen data and is capable of adapting to different domains with minor adjustments to training strategies.

CHAPTER 7

CONCLUSION AND FUTURE ENHANCEMENTS

7.1 CONCLUSION

The proposed approach in this study involves an end-to-end fully convolutional network aimed at directly modeling the intricate nonlinear mapping between face photos and sketches. Experimental findings underscore the efficacy of the fully convolutional network in adeptly addressing this challenging task, facilitating pixel-wise predictions with both effectiveness and efficiency.

7.2 FUTURE ENHANCEMENTS

Future enhancements will focus on refining the existing loss function and conducting experiments across various databases. Additionally, investigations into the correlation between our approach and non-photorealistic rendering methodologies will be pursued.

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APPENDIX

A. SAMPLE CODE

```
import re
import os
from tqdm import tqdm
import numpy as np
import cv2
import matplotlib.pyplot as plt
import seaborn as sns
import tensorflow as tf
import tensorflow.keras
from tensorflow.keras.models import Model, Sequential
from tensorflow.keras.preprocessing.image import img_to_array
from tensorflow.keras.layers import Dense, Conv2D, MaxPool2D, UpSampling2D, Dropout, Input, Conv2DTranspose, BatchNormalization
from tensorflow.keras.optimizers import Adam, RMSprop
from tensorflow.keras.layers import LeakyReLU
```

```
from tensorflow.keras import backend as K
optimizer = RMSprop(lr=0.00005)
```

```
WARNING:absl:`lr` is deprecated, please use `learning_rate` instead, or use the legacy optimizer, e.g.,tf.keras.optimizers.legacy.RMSprop.
```

```
smooth = 1.

def discriminator_loss(y_true,y_pred):
    BATCH_SIZE=30
    return K.mean(K.binary_crossentropy(K.flatten(y_pred), K.concatenate([K.ones_like(K.flatten(y_pred[:BATCH_SIZE])),K.zeros_like(K.flatten(y_pred[:BATCH_SIZE

def discriminator_on_generator_loss(y_true,y_pred):
    BATCH_SIZE=30
    return K.mean(K.binary_crossentropy(K.flatten(y_pred), K.ones_like(K.flatten(y_pred))), axis=-1)

def generator_l1_loss(y_true,y_pred):
    BATCH_SIZE=30
    return K.mean(K.abs(K.flatten(y_pred) - K.flatten(y_true)), axis=-1)

def least_absolute_error(y_true, y_pred):
    return (K.abs(y_pred - y_true))

def wasserstein_loss(y_true, y_pred):
    return K.mean(y_true * y_pred)

def mean_squared_error(y_true, y_pred):
    return 10*K.mean(K.square(y_pred - y_true), axis=-1)
```

```python
image_path = 'Dataset/photos'
img_array = []
```

```python
sketch_path = 'Dataset/sketches'
sketch_array = []
```

```python
SIZE = 256
```

```python
def sorted_alphanumeric(data):
    convert = lambda text: int(text) if text.isdigit() else text.lower()
    alphanum_key = lambda key: [convert(c) for c in re.split('([0-9]+)',key)]
    return sorted(data,key = alphanum_key)
```

```python
sketch_file = sorted_alphanumeric(os.listdir(sketch_path))
```

```python
image_file = sorted_alphanumeric(os.listdir(image_path))
```

```python
for i in tqdm(image_file):
    image = cv2.imread(image_path + '/' + i,1)
    image = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)
    image = cv2.resize(image, (SIZE, SIZE))
    image = image.astype('float32') / 255.0
    img_array.append(img_to_array(image))
    img1 = cv2.flip(image,1)
    img_array.append(img_to_array(img1))
    img2 = cv2.flip(image,-1)
    img_array.append(img_to_array(img2))
    img3 = cv2.flip(image,-1)
    img3 = cv2.flip(img3,1)
    img_array.append(img_to_array(img3))
    img4 = cv2.rotate(image, cv2.ROTATE_90_CLOCKWISE)
    img_array.append(img_to_array(img4))
    img5 = cv2.flip(img4,1)
    img_array.append(img_to_array(img5))
    img6 = cv2.rotate(image, cv2.ROTATE_90_COUNTERCLOCKWISE)
    img_array.append(img_to_array(img6))
    img7 = cv2.flip(img6,1)
    img_array.append(img_to_array(img7))
```

```
100%|████████████████████████████████████████████████████████████| 188/188 [00:02<00:00, 86.38it/s]
```

```python
print("Total number of sketch images:",len(sketch_array))
print("Total number of images:",len(img_array))
```

```
Total number of sketch images: 1504
Total number of images: 1504
```

```python
def plot_images(image, sketches):
    plt.figure(figsize=(7,7))
    plt.subplot(1,2,1)
    plt.title('Sketches ', color = 'black', fontsize = 20)
    plt.imshow(sketches)
    plt.subplot(1,2,2)
    plt.title('Image', color = 'green', fontsize = 20)
    plt.imshow(image)


    plt.show()
```

```python
ls = [i for i in range(0,65,8)]
for i in ls:
    plot_images(img_array[i],sketch_array[i])
```

```
train_sketch_image = sketch_array[:1000]
train_image = img_array[:1000]
```

```
test_sketch_image = sketch_array[1000:]
test_image = img_array[1000:]
```

```
train_sketch_image = np.reshape(train_sketch_image,(len(train_sketch_image),SIZE,SIZE,3))
```

```
train_image = np.reshape(train_image, (len(train_image),SIZE,SIZE,3))
```

```
print('Train color image shape:',train_image.shape)
```

```
Train color image shape: (1000, 256, 256, 3)
```

```
test_sketch_image = np.reshape(test_sketch_image,(len(test_sketch_image),SIZE,SIZE,3))
```

```
test_image = np.reshape(test_image, (len(test_image),SIZE,SIZE,3))
```

```
print('Test color image shape',test_image.shape)
```

```
Test color image shape (504, 256, 256, 3)
```

```
def model():
    encoder_input = Input(shape = (SIZE, SIZE, 3))
    x = downsample(16, 4, False)(encoder_input)
    x = downsample(32,4)(x)
    x = downsample(64,4,False)(x)
    x = downsample(128,4)(x)
    x = downsample(256,4)(x)

    encoder_output = downsample(512,4)(x)

    decoder_input = upsample(512,4,True)(encoder_output)
    x = upsample(256,4,False)(decoder_input)
    x = upsample(128,4, True)(x)
    x = upsample(64,4)(x)
    x = upsample(32,4)(x)
    x = upsample(16,4)(x)
    x = Conv2DTranspose(8,(2,2),strides = (1,1), padding = 'valid')(x)
    decoder_output = Conv2DTranspose(3,(2,2),strides = (1,1), padding = 'valid')(x)
    return Model(encoder_input, decoder_output)
```

```
model.compile(optimizer = Adam(learning_rate = 0.001), loss = 'mean_absolute_error',
              metrics = ['acc'])
```

```
model.fit(train_image, train_sketch_image, epochs = 100, verbose = 1)
```

```
prediction_on_test_data = model.evaluate(test_image, test_sketch_image)
print("Loss: ", prediction_on_test_data[0])
print("Accuracy: ", np.round(prediction_on_test_data[1] * 100,1))
```

B. SCREENSHOTS:

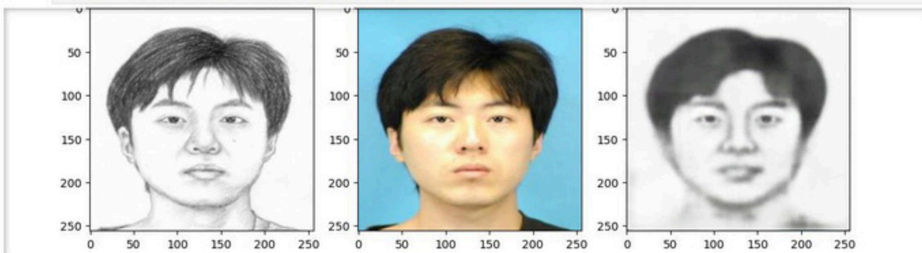
```
[68]: def show_images(real,sketch, predicted):
          plt.figure(figsize = (12,12))
          plt.subplot(1,3,1)
          plt.title("sketch",fontsize = 15, color = 'Blue')

          plt.imshow(sketch)
          plt.subplot(1,3,2)
          plt.title("Image",fontsize = 15, color = 'Lime')

          plt.imshow(real)
          plt.subplot(1,3,3)
          plt.title("Predicted",fontsize = 15, color = 'gold')

          plt.imshow(predicted)
```

```
[69]: ls = [i for i in range(0,95,8)]
      for i in ls:
          predicted =np.clip(model.predict(test_image[i].reshape(1,SIZE,SIZE,3)),0.0,1.0).reshape(SIZE,SIZE,3)
          show_images(test_image[i],test_sketch_image[i],predicted)
```



B.1 Model Prediction And Sketch Matched

C.PUBLICATION DETAILS

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1 message

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in the "14th International Conference on Science & Innovative Engineering" held on 27th & 28th April 2024 at Prince Dr. K. Vasudevan College Of Engineering & Technology

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